Lab #6 - Transverse Modes
ECE 482/582, Fall 2003

Goals:

1. Learn and demonstrate proper alignment techniques for a conventional cavity laser.
2. Force a HeNe laser into many different transverse modes by altering the gain in various places.
3. Study the mode intensity profiles of a HeNe laser.
4. Demonstrate how to polarize a laser using an internal Brewster angle window in the cavity

Reading:

Kuhn, *Laser Engineering*, Chapter 4: section 4.1 – 4.3, Appendix A.3

Other Resources:

Kogelnik and Li, Laser Beams and Resonators, Proc. IEEE, vol. 54, no.10, pp. 1312-1329, Oct.1966

Ramo, *Fields and Waves in Communication Electronics*, Chapter 14

Java applet - www.iap.uni-bonn.de/lehre/ss01_laserphysics/optres.html

Preparatory Questions:

1. What is Brewster's angle for HeNe at 632.8 nm and a pyrex cover slip, n ~ 1.5?

2. Explain why the paraxial approximation is used when solving the wave equation.

3. (Extra Credit) Use Maple, Mathematica or another tool to plot the cylindrical modes given by equation 4.18.

Laboratory Experiment:

You will be working with a modified HeNe laser in an effort to see the transverse modes of the laser cavity. Be careful with the setup. Many parts are fragile and high voltage is involved. **Do not touch the red lead from the power supply to the laser gain tube. It will be at ~1200Volts DC when operating and could be lethal if touched while grounded.**

Procedure:

1. Using the post mounted iris align the “guide” HeNe laser output to be directly along the optical rail and parallel to the tabletop.
2. Mount the laser gain tube inside its protective Lexan cylinder onto the far end of the optical rail positioning the front surface of the 100% reflecting flat mirror (on the anode or + HVend of the plasma tube) directly at the end of the rail at position 0.0 cm. [Then the rail readings correspond directly to the laser cavity length.]
3. Using the 3 nylon screws on both ends of the Lexan tube, <u>carefully</u> adjust the position of the laser plasma tube (without the power on to the plasma) to be perfectly coaxial to the “guide” laser beam. This is the most critical, and tedious, adjustment of the process. You will have to dim the lights a bit and look carefully at the reflection from the 100% mirror coming back to the alignment laser and also at the small amount of light transmitted through the “100% mirror.” When properly aligned, the reflected laser beam should go directly back into the HeNe alignment laser and there should be a very dim but clear red spot centered within a very dim circular ring from the far end of the tube. At this point the plasma gain tube is coaxial with the alignment laser beam and parallel to the rail.
4. Now place the laser output mirror (radius of curvature = 45 cm, reflectivity = 99% at 632.8 nm) on the rail with the front surface of the mirror positioned about 40-43 cm from the 0.0 cm position. Since the front surface of the mirror (the curved, reflecting surface) reflects 99% of the HeNe laser light and the back surface is antireflection (AR) coated, you will see only a bright reflected spot from the front surface reflecting back toward the alignment laser. Use the xy-positioners on the laser mirror mount to reflect the beam directly back into the alignment laser. At this point, the two laser mirrors on the laser should be very close to aligned.
5. Now block the alignment laser beam with a piece of cardboard, paper, etc. but be careful not to bump this laser. Turn on the power supply to the laser gain tube by switching the switch on the grey metal box. **<u>Be careful not to touch the red lead.</u>** At this point, your laser may very well begin lasing as evidenced by a bright red spot on a paper in front of the alignment laser. If not, try “rastering” the xy adjustment knobs on the output mirror mount by first turning the “x” knob back and forth by a small amount coming back to its initial position. If you see no lasing, then turn the “y” knob a small amount one direction (Remember which way you turned it and about how much!) and repeat the “x” back and forth adjustment. If no lasing, turn “y” a bit more in the same direction and repeat the “x” rocking. If no lasing, go back to the original “y” position and try a small turn the other direction. If, after “walking” the beam around, you still see no lasing, turn on (or unblock) the HeNe alignment laser and check your alignment again.
6. Once you get the laser lasing, adjust “x” and “y” to maximize the laser output intensity. Then turn off the alignment laser and move it out of the way, place the lens on the rail in the output beam and expand the beam onto the wall to show the transverse mode structure. Tweak the “x” and “y” knobs to try getting several different transverse lasing modes. Sketch several of the modes into your lab notebook. Keep adjusting the mirror to see if you can get several different “pure”

Hermite-Gaussian modes (or maybe even some cylindrical modes). Identify these modes by number and sketch them into your notebook, too. Measure the maximum power level you get from the laser.

7. Check the polarization characteristics of the modes you are making by placing a linear polarizer into the beam and rotating it 360 degrees. Are they polarized?
8. Now place the microscope cover slip into the beam inside the cavity, ie, between the laser gain tube window and the output mirror, and adjust it to Brewster's angle. (You might have to move the glass window in or out a bit to find a low-loss spot that will lase.) Now check the polarization of the laser mode. Is it polarized now? Did the transverse mode profile change with the polarizer? Why or why not?
9. Remove the Brewster's angle window and place the small aperture inside the cavity. Try to force the laser into the TEMoo mode by making the aperture smaller. You will have to tweak the mirror adjustment to make sure the mode is going right through the center of the iris.
10. For the TEMoo mode use the Beam Scan ("signal out" to scope) to measure the beam intensity profile in the two main perpendicular directions. How do your measured signals compare with what you would expect? You can also verify that the higher order modes are all larger than the TEMoo mode.
11. Remove the iris and put the rotating stage with the small, sharp needle mounted into the center into the cavity with the needle tip barely into the laser beam. Use this "forced zero" in the electric field (and, hence, the intensity pattern) to try forcing certain higher order modes to lase. Try rotating the needle and forcing the transverse mode pattern to also rotate.
12. Remove all intra-cavity items (needle, Brewster's angle glass, iris) and maximize the laser output. Now move the output mirror out 1-3cm to make a longer cavity. If the beam goes out, tweak the xy mirror adjustments to make it lase again. Keep moving to longer cavity lengths until you can no longer make the laser lase by adjusting the mirror. At what length does lasing stop?

<u>Questions to Answer:</u>

- We didn't adjust the cavity length to any particular length, especially to orders of a half wavelength. Why does the laser still lase for this arbitrary length?
- Why does the laser not laser if the output mirror is put at more than 45 cm?
- Why do the sizes of the laser modes get larger for longer cavities and smaller for shorter cavities?
- If the output mirror is at 40 cm, what is w_o, radius of curvature at the output mirror, z_o, and the beam waist at 1 m from the output mirror for TEMoo mode?
- How does the Brewster angle cover slip cause the laser to be polarized?
- Why does an iris force the laser to lase in the TEMoo mode?
- What is the effect of the needle tip into the beam? Why?

<u>Conclusion:</u>

You have aligned a gain cavity to get it to lase and observed the transverse mode structure. The affects of polarization and apertures on the transverse mode structure were investigated.